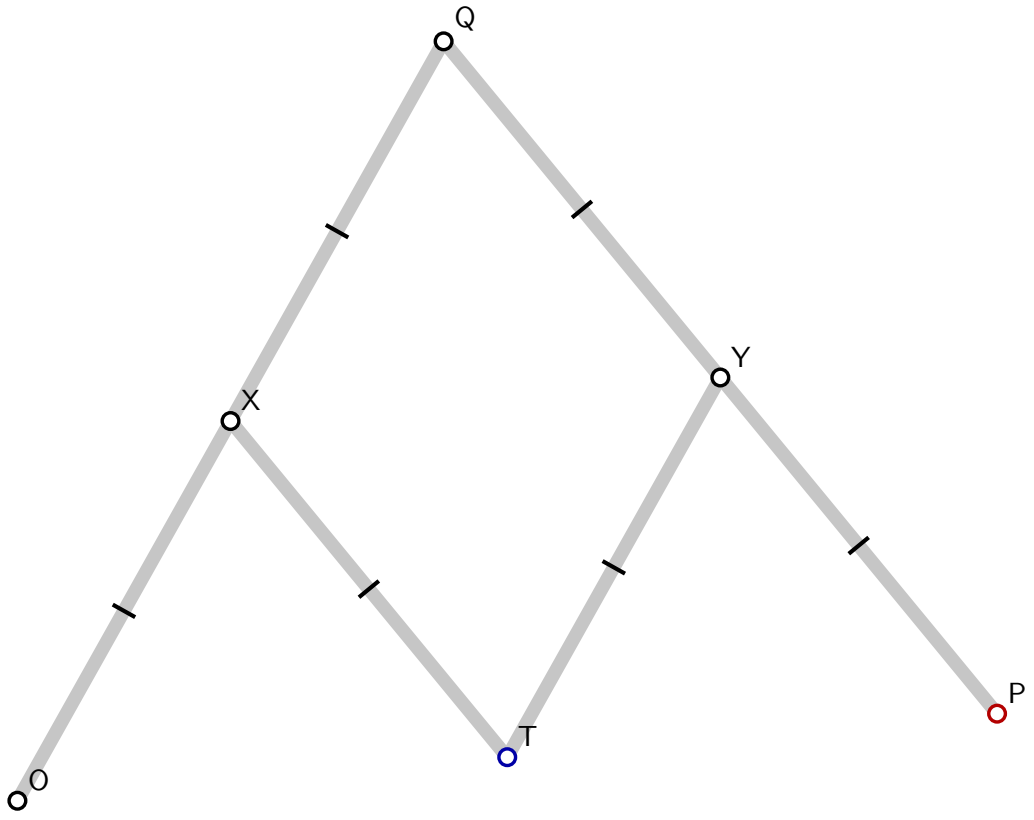
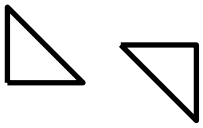
Keep O fixed. Move T; P draws. Keep the four-sided frame open and uncrossed. Each new copy goes on a fresh overlay. An adult moves the anchor or swaps the tips when a problem calls for it.

Problem 1: The six marked lengths are equal. Must P always be twice as far from O as T? Explain using the joined bars.



One square piece is one square unit. Two matching triangle pieces make one square unit.

Two halves



Join edges

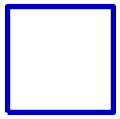


1 square unit

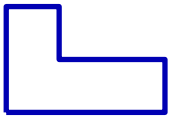


Problem 2: Use inserts G, H and I with the square and triangle pieces. How many square units cover each original and its copy? Find a rule for all three.

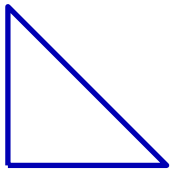
G



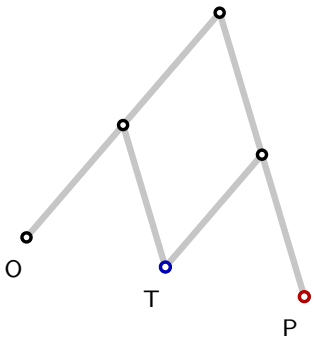
H



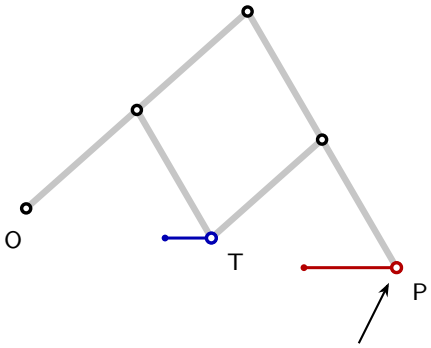
I



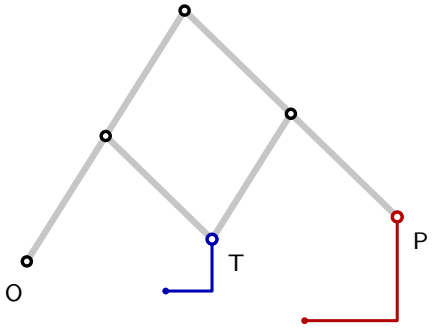
Start



Move P



Copy

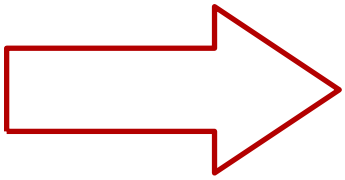


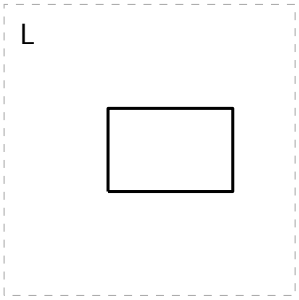
Problem 3: Now P traces and T draws. Trace J and K. What copies does T make? Find a rectangle for P that makes T follow the small solid box on insert L.

J



K





Problem 4: Trace triangle M with anchor O, then with anchor O′, keeping M in the same place. What changes in the two copies? Find a new place for M that makes the second copy land exactly on the first.

